dubbed “matrix mechanics” after the unfamiliar mathematical objects at its heart. Heisenberg’s matrix mechanics was technically forbidding and impossible to visualize—but it offered the prospect of a theory not just for atomic spectra but for the entire quantum world.

Einstein had started his own revolution in physics twenty years earlier, when he was Heisenberg’s age—and in a kind of isolation as well, though it wasn’t brought on by hay fever. In 1905, while working as a patent clerk in Switzerland, Einstein published his theory of special relativity, resolving a long-standing debate over the nature of light. Before Einstein, light was thought to be a wave in some kind of as-yet-undetected medium with the (spectacularly nineteenth-century) name of *luminiferous aether.* But in 1887, physicists Albert Michelson and Edward Morley had attempted to detect the motion of the Earth through the aether, and failed. Increasingly complex ad hoc ideas were thrown around to account for the results of the experiment. One physicist suggested that the results could be a sign that the aether compressed objects as they moved through it. Another physicist pointed out that wouldn’t be enough—the aether would also have to slow down all physical processes in objects moving through it! Yet allowing the aether to have these strange properties, all while maintaining its insubstantial nature, was increasingly difficult to believe or understand.

Einstein resolved the confusion in a brilliant stroke, the kind that is obvious only in hindsight. The aether, he proposed, was difficult to imagine because it didn’t exist at all. Light was simply a wave of electromagnetic fields, with no medium necessary, always traveling at a constant speed. From that simple assumption, Einstein spun out an entire theory of motion, the theory of special relativity. Special relativity was able to account for the negative result of the Michelson-Morley experiment, and it derived from first principles all of the strange effects—length contraction, time dilation—that others had only been able to assume.

Special relativity also made novel predictions. One consequence of the theory was that the speed of light was an absolute speed limit: no object or signal could go faster than the speed of light in a vacuum. The mathematics